

Samhita Pal, Ph.D.

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Employment History

- June 2025 – Present  **Postdoctoral Research Associate in Biostatistics**, Vanderbilt University Medical Center.
Advisors: Dr. [Amir Asiaee](#) (VUMC) and Dr. [Jared Huling](#) (University of Minnesota)
- Aug 2024 – May 2025  **Graduate Teaching Assistant**, Department of Statistics, North Carolina State University.
- June 2024 – August 2024  **Summer Biometric Intern**. Oncology Department, AstraZeneca Pharmaceuticals.
- Jan 2022 – May 2024  **Graduate Research Assistant**, Department of Statistics, North Carolina State University.
- Jan 2021 – Dec 2021  **Graduate Teaching Assistant**, Department of Statistics, North Carolina State University.
- August 2019  **Graduate Program**, Workshop on Statistical Data Integration, Institute of Mathematical Sciences, National University of Singapore.
- May – July 2019  **Summer Research Internship**, Department of Mathematics and Statistics, Indian Institute of Science Education and Research, India.

Education

- Jan 2021 – May 2025  **Ph.D. Statistics, North Carolina State University**, Raleigh, North Carolina, USA.
Thesis title: [Bayesian Inference in High-dimensional Regression Models using Sparse Projection Posterior](#).
Advisor: [Subhashis Ghoshal](#). GPA: 4.00.
- Aug 2018 – Oct 2020  **M.Sc. Statistics, University of Calcutta**, Kolkata, India.
Course rank: 2.
- Jul 2015 – May 2018  **B.Sc. Statistics (Honours), St. Xavier's College (Autonomous)**, Kolkata, India.
Course rank: 1.

Research Interests

 Bayesian Methods | High-dimensional Statistics | Causal Inference | Transfer Learning |

Research Publications

Published Journal Articles

-  [Dhrubajyoti Ghosh](#) and [Samhita Pal](#), Efficacy analysis in clinical trials: A comprehensive review of statistical and machine learning approaches, *New England Journal of Statistics in Data Science*, 2026, in press.
-  [Samhita Pal](#), [Dhrubajyoti Ghosh](#), and [Shu Yang](#), Penalized fci for causal structure learning in a sparse dag for biomarker discovery in parkinson's disease, *Accepted at Annals of Applied Statistics*, 2026.

- 3 Samhita Pal and Subhashis Ghoshal, Bayesian high-dimensional grouped-regression using sparse projection-posterior, *Statistica Sinica*, 2026, in press.
- 4 Samhita Pal and Subhashis Ghoshal, Bayesian high-dimensional linear regression with sparse projection-posterior, *Accepted at Bayesian Analysis*, 2026.
- 5 Dhrubajyoti Ghosh, Samhita Pal, Michael Lutz, and Sheng Luo, Ensemble survival analysis for preclinical cognitive decline prediction in alzheimer's disease using longitudinal biomarkers, *Journal of Alzheimer's Disease*, vol. 107, no. 3, pp. 1256–1266, 2025.
- 6 Samhita Pal and Subhashis Ghosal, Projection-posterior for variable selection: Weak limit and coverage, *Electronic Journal of Statistics*, vol. 19, no. 2, pp. 3743–3770, 2025.
- 7 Samhita Pal and Xinge Jessie Jeng, Discovering candidate genes regulated by gwas signals in cis and trans, *Statistics in Biosciences*, 2024.

Published Conference Proceedings

- 1 Samhita Pal and Amir Asiaee, Improving rct-based treatment effect estimation under covariate mismatch via calibrated alignment, in *UAI 2026*.
- 2 Amir Asiaee, Samhita Pal, Cole Beck, and Jared D. Huling, Sharp bounds for treatment effect generalization under outcome distribution shift, in *5th Conference on Causal Learning and Reasoning (CLearR)*, 2026.

Revision Submitted

- 1 Amir Asiaee, Chiara Di Gravio, Cole Beck, Yuting Mei, Samhita Pal, and Jared D. Huling, Improving precision of rct-based cate estimation using data borrowing with double calibration, *Revision submitted to Journal of Machine Learning Research*, 2026.
- 2 Samhita Pal and Dhrubajyoti Ghosh, Network-aware IV regression for causal node discovery and estimation, *Revision submitted to Journal of the American Statistical Association*, 2026.
- 3 Samhita Pal and Xiaoling Wu, Alpha-allocation based on utility and likelihood in multi-arm trials, *Revision submitted to Biometrical Journal*, 2026.

Submitted

- 1 Dhrubajyoti Ghosh and Samhita Pal, Representation learning for causal inference in multimodal repeated cross-sectional surveys with block-missingness, *Submitted to Statistics in Medicine*, 2026+.
- 2 Amir Asiaee, Samhita Pal, James O'quinn, and James P. Long, Causal discovery with mixed latent confounding via precision decomposition, *Submitted*, 2026+.
- 3 Samhita Pal and Jared Huling, Heterogeneous effects of continuous treatments via conditional incremental effects, *Submitted to Journal of the American Statistical Association*, 2026+.
- 4 Samhita Pal, Amir Asiaee, and Jared Huling, Improving RCT-based CATE estimation under covariate mismatch via double calibration, *Submitted to Journal of the American Statistical Association*, 2026+.
- 5 Samhita Pal and Tian Gu, Hierarchical projection for adaptive knowledge transfer, *Submitted to Journal of the American Statistical Association*, 2026+.
- 6 Samhita Pal, Dilshad Imon, and Dhrubajyoti Ghosh, Air pollution and public health: A lagged spatio-temporal analysis of copd and stroke across u.s. cities, *Submitted to Journal of Data Science*, 2026+.

- 7 **Samhita Pal**, James O'quinn, Kaveh Aryan, Heather Pua, James P. Long, and **Amir Asiaee**, DAG DECORation: Continuous optimization for structure learning under hidden confounding, *Submitted*, 2026+.
- 8 Parvez Rashid, **Samhita Pal**, Vinay Kumar Samudrala, Rahul Yedida, and Edward Gehringer, A bayesian model for peer assessors' reputation system, *Submitted to EDM 2026*,

Working Papers

- 1 **Samhita Pal** and Nilanjan Chatterjee, Projection-posterior approaches for polygenic risk score estimation, *Working paper*, 2026+.
- 2 **Samhita Pal**, **Amir Asiaee**, and **Jared D. Huling**, CATE transport sensitivity analysis with doubly valid doubly sharp estimation, *Working paper*, 2026+.
- 3 **Samhita Pal** and **Chang Su**, Transfer learning for causal structure discovery across related DAGs, *Working paper*, 2026+.
- 4 **Samhita Pal** and **Subhashis Ghosal**, Bayesian sparse projection-posterior approach for high-dimensional generalized linear models, *Working paper*, 2026+.
- 5 Shubhajit Sen, **Samhita Pal**, and Srijan Sengupta, Social sensors in epidemiological networks via graph eigenvectors, *arXiv preprint arXiv:2112.14385*, 2021, Working paper.

Selected Collaborators

 **Subhashis Ghoshal** (NC State University) | **Amir Asiaee** (VUMC) | **Jared Huling** (University of Minnesota) | **Shu Yang** (NC State University) | **Dhrubajyoti Ghosh** (Duke / Kennesaw State University) | **Tian Gu** (Columbia University) | **Chang Su**

Skills

Research	 High-dimensional Models, Bayesian Modeling and Inference, Causal Inference, Instrumental Variable Regression, Transfer Learning.
Computing	 R (Proficient), sas (Intermediate), Python (Elementary), $\LaTeX$.
R Packages	 sparseProj, credInt, PFCI, eGene, ivgls — https://github.com/SamhitaPal3
Languages	 English, Bengali, Hindi, Spanish (elementary).

Miscellaneous

Honors & Awards

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| 2026 |  Honorable Mention (2nd place), ISBA Savage Award for dissertation, Theory and Methods category.
 Travel Award , IMSI New Horizons on Model Transportability and Data Integration. |
| 2024 |  Student Conference Award , NC State Graduate Student Association Award for Travel and Conferences.
 Student Paper Competition Travel Award , IMS ICSDS. |
| 2023 |  Travel Award , WSDS.
 National Science Foundation Travel Award , New Frontiers in Reliability and Risk Analysis Symposium.
 Student Paper Competition Travel Award , IISA. |

Miscellaneous (continued)

- 2022  **B.B. Bhattacharyya Graduate Fund for Excellence Award** (nominated 2022; won 2023).
- 2022  **Harshbarger Student Travel Award**, SRCOS Conference.
-  **Gertrude M. Cox Achievement Award** for Outstanding Ph.D. Qualifier Exam.
- 2015–2020  **Scholarship for Higher Education (SHE): INSPIRE**, Department of Science and Technology, Government of India.
- 2019  **Third Prize for Poster**, National Seminar for Sample Surveys, India.
- 2020  **Second Rank**, M.Sc. Statistics, University of Calcutta.
- 2018  **Arts and Science Award for First Rank**, B.Sc. Statistics, St. Xavier's College (Autonomous), Kolkata.
- 2015  **First Rank in school**, under-25 rank in state, Higher Secondary Examination, West Bengal, India.
- 2013  **First Rank in school**, under-25 rank in state, Secondary Examination, West Bengal, India.
-  **Honorary Student Membership**, Mu Sigma Rho.

Conference & Presentations

- 2026  **Finalist, ISBA Savage Award** : Best Dissertation in the Theory and Methods category.
- 2024  **Student Paper Award Session** : ICSDS: International Conference on Statistics and Data Science.
-  **Invited Talk** : CFE-CMStatistics: 18th International Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics.
-  **Invited Talk** : Calcutta Statistical Association Triennial Symposium.
-  **Poster Presentation** : Graduate Student Research Symposium, NCSU.
- 2023  **Speed Talk and Hybrid Poster Session** : WSDS: Women in Statistics and Data Science.
-  **Invited Poster Session** : New Frontiers in Reliability and Risk Analysis Symposium: A Tribute to Nozer D. Singpurwalla.
-  **Poster Presentation** : JSM: Joint Statistical Meetings.
-  **Student Paper Competition** (Theory and Probability division) : IISA: International Indian Statistical Association.
- 2022  **Poster Presentation** : SRCOS: Southern Regional Council on Statistics.
- 2019  **Poster Presentation** : National Seminar for Sample Survey, University of Calcutta.

Teaching Experience

- Fall 2024  **Grader**, ST 558 — Data Science for Statisticians, NC State University.
-  **Grader**, ST 563 — Introduction to Statistical Learning, NC State University.
- Summer 2023  **Grader**, ST 503 — Fundamentals of Linear Models and Regression, NC State University.
- Fall 2021  **Grader**, ST 312 — Introduction to Statistics II, NC State University.
- Summer 2021  **Grader**, ST 311 — Introduction to Statistics, NC State University.
- Spring 2021  **Lab Instructor**, ST 312 — Introduction to Statistics II, NC State University.

Positions of Responsibility

- 2025  Organized Invited Session, “Advances in statistical modeling and machine learning for complex data,” 2025 CFE-CMStatistics.

Miscellaneous (continued)

- 2022–2024
- Founder President, Triangle Bengali Association, an NC State University organization for the Bengali-speaking community.
 - International Students Representative, Statistical Graduate Students Association, NC State University — organized 4 events.
 - Volunteer, Office of International Services, NC State University.

Paper Reviews

- Calcutta Statistical Association Bulletin; Annals of Statistics; Asian Research Journal of Mathematics; CLear Conference 2026; Bayesian Analysis; AISTATS CauScale 2026; BMC Medical Informatics and Decision Making; Frontiers in Applied Mathematics and Statistics; Asian Journal of Probability and Statistics; NeurIPS 2026.

Other Activities

- Completed up to the 4th year in Indian classical dance forms — Bharatnatyam and Manipuri — from Pracheen Kala Kendra, Chandigarh, India.
- Completed up to the 4th year in Fine Arts from Sarbabharatiya Sangeet-O-Sanskriti Parishad, recognised by the West Bengal State Akademy of Dance, Drama, Music, Fine Arts and Raja Mansingh Tomar Music Arts University, Gwalior, MP.
- Ukulele, classical music, badminton.

References

Available on Request